

Krutika Dave

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[Linkedln](#)

Experience

GalaxEye Space

AI Engineer

Bangalore, India

03|2024 – Present

- Developed an **ensemble statistical vehicle detection model**, achieving **96% TPR, 12% FPR, and 4% FNR**; generalized on unseen proprietary droneborne SAR data [Manuscript under review]. Subsequently built a YOLO and DETR based target detection model.
- Led Development of a **multimodal Optical–SAR anomaly detection model**, achieving **>80% mIoU** on unseen scenes; deployed as part of the **company’s core defence product**, and used by clients.
- R&D team: Currently researching and developing state-of-the-art foundation **joint embedding architecture** for multimodal remote sensing data.

Geospatial Analyst Intern

10|2023 – 02|2024

Streamlined Proof-of-Concept execution with aquaculture companies, collaborating across sales, science, and product engineering team. Built and successfully delivered a client PoC, assessing one-year changes in Lake Mar Chiquita, Argentina, using Sentinel-2 data and NDWI-based analysis in GEE.

Space Applications Center, ISRO | Mentor - Dr. Sushil Kumar Singh

Ahmedabad, India

Research Intern

11|2022 – 05|2023

“Understanding Lake Ice Phenology of Mansarovar Lake using Remote Sensing from 2010-2022”

Studied Lake Ice Phenology (LIP) of Mansarovar Lake (2010–2022), defining six freeze–thaw variables based on date of occurrence. Leveraged **NASA MEaSUREs EASE-Grid 2.0 Tb ESDR passive microwave data and ERA5 air temperature data**. Decadal Study using **statistical Analysis**, further validated with **Landsat-8 and Sentinel-2** optical imagery. [Thesis link]

Arvind Limited

Ahmedabad, India

GIS Intern

06|2022 - 07|2022

Conducted hydrological analysis of Rajkot district using Cartosat-1 DEM for terrain and stream mapping, Sentinel-2 for soil and LULC mapping, and SWAT to delineate 21 sub-basins and map aquifers, floodplains, and HRUs. [Link].

Education

Ahmedabad University

Ahmedabad, India

Major - Integrated Masters in Life Science, CGPA- 3.31/4.00

2018-2023

[Relevant Coursework: Mathematics for Biological Sciences, Biophysics, Basics of Bioinformatics, Bioinformatics, Research Methodology, Energy and Climate change, Environmental Science, Ecology and Society]

Research Projects

Ahmedabad University

“Spatio-temporal changes in eco-geomorphological classes of coral reefs of Narara Island in the Gulf of Kutch with respect to sea surface temperature from 2000-2020” | Mentors - Prof. Ratna Ghosal & Dr. Anjali Bahuguna

Analysed Landsat-5 TM, Landsat-8 OLI/TIRS, and IRS LISS-III data using Maximum Likelihood and SVM classification (accuracy: 85–89% and 75–83%). Conducted change detection revealing increased mud/sand deposition. Integrated MODIS AQUA SST analysis over 20 years.

“Lakes of Ahmedabad and SDGs” | Mentor - Prof. Minal Pathak | Team Size: 5

Surveyed three urban lakes on cleanliness, biodiversity, and conservation measures via stakeholder interviews and field visits.

Produced a comparative report and short film linking findings to the UN Sustainable Development Goals.

Achievements

INSPIRE Scholarship (2018–2023) – 50% sponsorship for undergraduate studies; awarded for ranking in the top 1% in Class 12 Board Examination.

Newsletter Publication – Gujarat Science Academy: Co-authored "Is Nal Sarovar Dying?".

Featured Project | Ahmedabad University Expo – Created impactful visuals with voice-overs on key environmental issues, including ice melt, biodiversity loss, and poaching [\(link\)](#).

Leadership positions

- **AI Module Lead | Professional | 2026–Present:** [Driving an AI module from R&D to production.](#)
 - Cultural head | Chiasma'22: Managed a team of 30. Negotiated with various vendors for sponsorship deals.
 - Head of 'The Identity Hint' (Forensic Science) workshop | Chiasma'21: Designed the module. Formed a team that was trained to conduct a 3-day national workshop.
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Skills

Tools: ArcGIS, QGIS, SNAP, Google Earth Engine; **Technical Areas:** AI/ML Fundamentals, Multimodal Learning, Self-Supervised Learning (SSL); **Programming Languages:** Python
